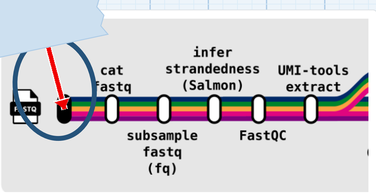
gxwf: Galaxy Workflows, Typed, Not Guessed

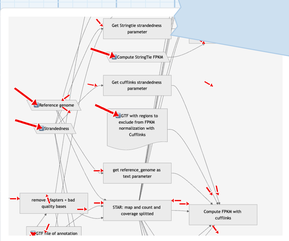
The Galaxy workflow command line: validate, lint, convert, clean
Docs: jmchilton.github.io/galaxy-tool-util-ts

Galaxy PROJECT

gxwf type-checks tool calls and connections before any job runs.



nf-core/rnaseq: **one** arrow



Galaxy: **arrows everywhere**

1 Most checks stop at the entrance

nf-schema checks the pipeline's own parameters (circled); flags inside each process are free text, as in Snakemake and WDL. In Galaxy, gxwf can type each tool call and each connection.

2 Format 2 makes the types writable

Readable YAML, so good for humans and agents alike.

.ga 1149 characters, one string

```
"tool_state": "{\"alignment\":
{\"__class__\": \"Connected …
```

Format 2 same step, excerpt

```
state:
  pe_parameters:
    paired_end_status: PE_fragments
```

A TYPO IN FORMAT 2

```
IWC chipseq-sr, MACS2 step: format: BAMX
$ gxwf validate chipseq-sr.gxwf.yml
   --connections --strict
FAIL macs2_callpeak · format
   Expected "BAM", actual "BAMX"
   Expected "BAMPE", actual "BAMX"
   Expected "BED", actual "BAMX"
   + 1 more union-branch line
unplanted file: 7 validated, 14 ok
```

TYPE-CHECK RECEIPT

IWC rnaseq-pe @ 23ccd9915 · gxwf 1.15.0

```
$ gxwf validate rnaseq-pe.ga --connections
   --strict-structure --strict-state
```

TOOL CALLS

	QTY	STATE
fastp 1.4.0+galaxy0	1	PASS
rna_star 2.7.11b+galaxy1	1	PASS
featurecounts 2.1.1+galaxy1	1	PASS
… 7 more tools	12	PASS
inside 4 subworkflows	24	PASS

TYPED

Tool state: 39 validated, 0 skipped

CONNECTIONS

	top level
list:paired → fastp	map over list
map-overs inferred	5
reductions into MultiQC	13

Connections: 57 ok, 0 invalid

WALL TIME < 1 s

warm cache, 5 runs	0.54–0.68 s
jobs run to reach this verdict	0

ALL OF IWC (tool state only)

133 workflows · tool calls validated	3,217
failed	0
skipped: version mismatch in file	1

TEAR HERE

VOID one typo planted in each copy

FAIL **featureCounts · paired_end_status**
Expected "PE_individual" | "PE_fragments", actual **"PE_fragment"**

INVALID **fastp → STAR · paired input**
output is collection<list>, input expects collection<paired>

VOID

REPRINT

one letter restored: paired_end_status: PE_fragments
featurecounts re-run, 3 runs 0.60–0.65 s
Tool state: 39 validated, 0 failed

PASS

END OF RECEIPT

3 Types from the Tool Shed

It serves typed schemas for 10,000+ community tools: names, types, select options, conditionals, collections.

4 Connections carry collection algebra

A list of pairs into a paired input: a map-over, inferred. A flat list into a paired input: rejected.

5 One validator, everywhere

The CLI and the gxwf-web browser app run one TypeScript core; the Galaxy Workflows VS Code extension's main branch reuses its tool-state checks. Under a second, so an agent can repair in a tight loop.

6 Runs in CI

No Python, no Galaxy server. Fill the tool cache, then validate-tree checks a whole repository's tool state, as for all of IWC on the receipt. Any failure exits non-zero, with a JSON or Markdown report for the log.

7 Caught before compute

A one-letter typo fails before any job is queued. Restore it and the re-run passes in under a second.

Where gxwf fits

gxwf lives in the inner loop of writing a workflow, by hand or by an agent. A check needs no Galaxy server and takes under a second on a warm cache, so it can run after every edit. Only a workflow that passes moves on to real tests.

draft
`Format 2 · .ga`
by hand, in an editor, or by an agent

→

↻ until it passes, under a second a lap

check
`gxwf validate`
each tool call's state, then the wiring

→

fix
the error names the parameter and what it allows

→

test
`validate-tests · Planemo`
real data, real jobs on Galaxy

One example. In the Galaxy Workflow Foundry, an agent drafts Format 2 and runs gxwf after each step it writes.

Types aren't biology: a type check can't tell that a spliced aligner is the wrong tool for a bacterial genome. A workflow that validates still needs real tests and a person's review.

GET STARTED

Install (Node.js)
```
npm install -g @galaxy-tool-util/cli
```
Cache the tool schemas it uses
```
galaxy-tool-cache populate-workflow
  my-wf.gxwf.yml
```
Check it, wiring included
```
gxwf validate my-wf.gxwf.yml
  --connections --strict
```
Check a whole repository (tool state)
```
gxwf validate-tree workflows/
  --strict-state
```
Turn a .ga into Format 2
```
gxwf convert my-wf.ga --to format2
```

Metro map: nf-core/rnaseq, CC BY; arrow added. Galaxy graph: Galaxy workflow editor; arrows added.
Workflows: github.com/galaxyproject/iwc @ 23ccd9915. Validator: npm @galaxy-tool-util/cli 1.15.0.
Source: github.com/jmchilton/galaxy-tool-util-ts